

A. ANALYST INFORMATION

- Analyst Name: **ODORI SHALOM ARIBITON**
- Date/Time Received: 25th June 2026 05:25AM
- Date/Time Analysis: 29th June 2026
- Date/Time Analysis Completed: 3rd JULY 2026 1:05AM

B. LOG FILE INFORMATION

- File Name: access-1.log
- File Type: ASCII text
- File Size: 25.6KB
- Source System:
- Source IP Address:
- Log Location: /home/kali/Downloads/access-1.log
- Hash Value : **SHA256hash**

19143051a4ca34453331da849f35f511dae543181b6d7a1312156932cfbe9c08

- Case/Incident Number:

C. INITIAL OBSERVATIONS

- Number of Lines: 156
- Log Format: Combined Log Format (Apache/Nginx)
- Delimiter Used: Double quote (")
- Time Range Covered: 23 hours and 10 minutes
- Primary Fields Identified: 9 fields which are listed below:
 - ❖ Client IP Address
 - ❖ Remote Logname
 - ❖ Authenticated User
 - ❖ Timestamp
 - ❖ Request Line (Method, Path, Protocol)
 - ❖ HTTP Status Code
 - ❖ Response Size
 - ❖ Referer
 - ❖ User-Agent

D. ANALYSIS OBJECTIVES

What am i looking for?

- ☒ Failed Logins
- ☒ Unauthorised Access
- ☒ Unique IP and request profiling
- ☒ Network Scanning
- ☒ Web Attacks
- ☐ Data Exfiltration
- ☐ Status code and response analysis

E. COMMANDS EXECUTED

Command	Purpose	Result
head -n 1 access-1.log	To verify the log format	Combined Log Format (Apache/Nginx)
wc -l access-1.log	To count the number lines	156
ls -lh access-1.log	Gives the file size	25.6KB
realpath access-1.log	Show the file location	/home/kali/Downloads/access-1.log
file access-1.log	Show file type	ASCII text
head -n 1 apache_logs grep -oP '\[.*?\]'	Time range started	Start : 02/May/2020 00:05:32 +0000
tail -n 1 access-1.log grep -oP '\[.*?\]'	Time range ended	End : 02/May/2020 23:15:37 +0000
grep -c "GET" access-1.log	Review GET request	127 entries found
awk -F'"' '{print \$3}' access-1.log awk '{print \$1}' sort -u wc -l	Uniques status codes	4 unique status codes
grep -c "CONNECT" access-1.log	Review CONNECT request	10 entries found
awk '{print \$9}' access-1.log sort uniq -c	Status codes	4

sort -rn		
LC_ALL=C grep -cP '\x[0-9A-Fa-f]{2}' access-1.log		11
awk -F'"' '{print \$6}' access-1.log grep -vE '^-?\$' sort -u wc -l	Unique user agents	55
grep -ic 'firefox' access-1.log	Request made by firefox	11 entries found

1. How many GET requests were logged in?

```
(kali@kali)-[~/Downloads]
$ grep -c 'GET' access-1.log
127
```

2. How many unique status codes were returned by the server

```
(kali@kali)-[~/Downloads]
$ awk -F'"' '{print $3}' access-1.log | awk '{print $1}' | sort -u | wc -l
4
```

3. How many HTTP tunneling attempts were made?

```
(kali@kali)-[~/Downloads]
$ grep -c 'CONNECT' access-1.log
10
```

4. Note the status codes and detail what they mean

```
(kali㉿kali)-[~/Downloads]
$ awk '{print $9}' access-1.log | sort | uniq -c | sort -rn
103 200
22 400
18 404
10 "-"
2 304
1 163
```

CODE	COUNT	MEANING
200	103	OK - request succeeded, content returned
400	33	Bad Request - malformed or invalid request syntax
404	18	Not found - path doesn't exit on the server
304	2	Not modified - client cache is still valid

5. How many entries have completely invalid request lines containing raw binary data?

```
(kali㉿kali)-[~/Downloads]
$ LC_ALL=C grep -cP '\\x[0-9A-Fa-f]{2}' access-1.log
11
```

6. Common reasons you may see invalid/binary request lines

These show up in access logs when:

1. **TLS/HTTPS ClientHello on a plain HTTP port** — a client tries to negotiate an SSL/TLS connection (`\x16\x03\x01` bytes) on a port expecting plain HTTP. The server can't parse it as HTTP so logs the raw bytes and returns 400. This is the most common cause — seen 10 times in this file from `44.224.22.196`
2. **SOCKS proxy probing** — tools testing if the server will act as a SOCKS4 (`\x04\x01`) or SOCKS5 (`\x05\x01`) proxy send binary handshake bytes instead of HTTP
3. **Port scanners & vulnerability scanners** — tools like Shodan, Masscan, or Nmap send raw payloads to fingerprint services, not caring whether HTTP is expected
4. **Protocol confusion** — a client misconfigured to send the wrong protocol (e.g. a database client hitting a web server port)
5. **Fuzzing/random payloads** — automated tools deliberately sending garbage data to test how the server handles unexpected input

7. How many unique user agents were observed, excluding empty or missing user agents

```
(kali@kali)-[~/Downloads]
$ awk -F'"' '{print $6}' access-1.log | grep -vE '^-?$' | sort -u | wc -l
55
```

8. How many request were made by firefox

```
(kali@kali)-[~/Downloads]
$ grep -ic 'firefox' access-1.log
11
```

F. FINDINGS:

1. Ten HTTP tunneling attempts via CONNECT requests to `54.236.11.149:80` were detected, all rejected with status 400

G. INDICATOR OF COMPROMISE (IoCs)

Suspicious IP Addresses: 44.224.22.196 and 66.240.205.34

```
(kali㉿kali)-[~/Downloads]
$ grep -E 'CONNECT|169\.254\.169\.254|\\x' access-1.log | awk '{print $1}' | sort -u
44.224.22.196
66.240.205.34
```

Domains:

```
(kali㉿kali)-[~/Downloads]
$ grep -oP 'https?://[a-zA-Z0-9._/-]+' access-1.log | sort -u
http://169.254.169.254/
http://169.254.169.254/latest/dynamic/instance-identity/document
http://19ce033f.ngrok.io/arm7
http://54.236.11.149
http://54.236.11.149/
http://cloudsystemnetworks.com
http://code.google.com/appengine
http://example.com/
https://github.com/AliasIO/Wappalyzer
https://radioresidencias.com/di678bc/lat47na/
http://www.bing.com
http://www.bing.com/search
http://www.googlebot.com/bot.html
http://www.google.com/bot.html
```

```
(kali㉿kali)-[~/Downloads]
$ grep -oP '169\.254\.169\.254[a-zA-Z0-9._/-]+' access-1.log | sort -u
169.254.169.254/
169.254.169.254/latest/dynamic/instance-identity/document
```

User Accounts: /boaform/admin/formPing

```
(kali㉿kali)-[~/Downloads]
$ grep -oP '(?=<GET |POST )/[A-Za-z0-9._/-]*(?:login|admin|user|account|password|auth|wp-admin|phpmyadmin)[a-zA-Z0-9._/-]*' access-1.log | sort -u
/boaform/admin/formPing
```

File Names: robots.txt and favicon.ico

```
(kali㉿kali)-[~/Downloads]
$ grep -oP 'GET \K[^\s]+' access-1.log | grep -oP '^[^/]+\.[A-Za-Z0-9]+$' | sort -u
favicon.ico
robots.txt
```

LOG FILE INFORMATION

- File Name:access-2.log
- File Type: ASCII text
- File Size: 14KB
- Source System:
- Source IP Address:
- Log Location: /home/kali/Downloads/access-2.log
- Hash Value : **SHA256hash**

8548404eda854a2ca3dc193f6b3dfae0ecbe3f59193b2311498ec0a0a5816ccf

C. INITIAL OBSERVATIONS

- Number of Lines:100
- Log Format: Combined Log Format (Apache/Nginx)
- Delimiter Used: Double quote (")
- Time Range Covered: 4 days 15 hours
- Primary Fields Identified:9 fields which are listed below:
 - ❖ Client IP Address
 - ❖ Remote Logname
 - ❖ Authenticated User
 - ❖ Timestamp
 - ❖ Request Line (Method, Path, Protocol)
 - ❖ HTTP Status Code
 - ❖ Response Size
 - ❖ Referer
 - ❖ User-Agent

D. ANALYSIS OBJECTIVES

What am i looking for?

- ☒ Failed Logins
- ☒ Unauthorised Access
- ☒ Status code and response analysis
- ☐ Network Scanning
- ☒ Web Attacks
- ☒ Malicious activity detection
- ☐ Other: Suspicious User Activity

E. COMMANDS EXECUTED

Command	Purpose	Result
head -n 1 access-2.log	To verify the log format	Combined Log Format (Apache/Nginx)
wc -l access-2.log	To count the number lines	100
ls -lh access-2.log	Gives the file size	14KB
realpath access-1.log	Show the file location	/home/kali/Downloads/access-2.log
head -n 1 access-2.log grep -oP '\[.*?\]'	Time range started	Start : 29/Sep/2015 21:15:18 -0400
tail -n 1 access-2.log grep -oP '\[.*?\]'	Time range ended	End: 03/Oct/2015 12:18:04 -0400
grep -c 'GET ' access-2.log	Review GET request	60 entries found
awk '{print \$1}' access-2.log sort -u wc -l	Unique IP's	48 unique IPs

awk -F'"' '{split(\$3,a," "); print a[1]}' access-2.log grep -c '^200\$'	Requests that yielded a 200 status	19
awk -F'"' '{split(\$3,a," "); print a[1]}' access-2.log grep -c '^400\$'	Requests that yielded a 400 status	38
grep -i 'doorbell\ dorbell' access-2.log	IP address range at the doorbell	186.64.69.141 /24
grep -io 'googlebot/[0-9.]*' access-2.log sort -u	Version of googlebot visited the website	Googlebot 2.1
grep -io 'firefox/[0-9.]*' access-2.log sort uniq -c sort -rn	Most frequent version of firefox	Firefox/31.0 (9 occurrences)
awk -F'"' '{print \$2}' access-2.log awk '{print \$1}' sort uniq -c sort -rn head -1	Most common HTTP method used	GET (60 occurrences)

1. How many GET requests were made?

```
(kali@kali)-[~/Downloads]
$ grep -c 'GET' access-2.log
60
```

2. How many unique IP addresses made requests?

```
(kali@kali)-[~/Downloads]
$ awk '{print $1}' access-2.log | sort -u | wc -l
48
```

4. How many requests yielded a 200 status

```
(kali@kali)-[~/Downloads]
$ awk -F'"' '{split($3,a," "); print a[1]}' access-2.log | grep -c '^200$'
19
```

5. How many requests yielded a 400 status

```
(kali㉿kali)-[~/Downloads]
$ awk -F'"' '{split($3,a," "); print a[1]}' access-2.log | grep -c '^400$'
38
```

6. What IP address range at the doorbell

```
(kali㉿kali)-[~/Downloads]
$ grep -i 'doorbell\\|dorbell' access-2.log
186.64.69.141 - - [30/Sep/2015:18:59:20 -0400] "GET /Ringing.at.your.dorbell!
HTTP/1.0" 404 162 "http://google.com/search?q=bitcoin" "x00_-gawa.sa.pilipin
as.2015"
```

7. What version of the googlebot visited the website?

```
(kali㉿kali)-[~/Downloads]
$ grep -io 'googlebot/[0-9.]*' access-2.log | sort -u
Googlebot/2.1
```

8. What was the most frequent version of firefox used for browsing website?

```
(kali㉿kali)-[~/Downloads]
$ grep -io 'firefox/[0-9.]*' access-2.log | sort | uniq -c | sort -rn
  9 Firefox/31.0
  4 Firefox/3.6.28
  2 Firefox/30.0
  1 Firefox/40.0
```

9. What is the most common HTTP method used?

```
(kali㉿kali)-[~/Downloads]
$ awk -F'"' '{print $2}' access-2.log | awk '{print $1}' | sort | uniq -c | s
ort -rn | head -1
 60 GET
```

F. FINDINGS

1. Twenty-one HTTP 404 errors were recorded indicating active directory and resource enumeration across multiple paths

G. INDICATOR OF COMPROMISE (IoCs)

Suspicious IP Addresses:

104.245.97.236, 137.117.108.84, 141.212.122.146 ,169.50.3.171 , 185.49.14.190
,185.25.151.159 , 186.64.69.141 , 193.19.118.8 , 198.7.58.194 , 80.82.70.24 , 169.50.3.171
,61.161.130.241 , 91.196.50.33 , 95.213.177.122 , 95.213.177.123 , 95.213.124 ,
95.213.177.125 , 95.213.177.126 , 185.25.151.159

```
(kali@kali) - [~/Downloads]
$ grep -E 'CONNECT|xmlrpc|CFIDE|manager/html|panel/.zip|admin|\\\\x|testproxy|dorbell|masscan|zgrab' access-2.log | awk '{pr
int $1}' | sort -u
104.245.97.236
137.117.108.84
141.212.122.146
141.212.122.202
169.50.3.171
185.25.151.159
185.49.14.190
186.64.69.141
193.19.118.8
198.7.58.194
58.213.123.107
58.55.121.17
61.161.130.241
80.82.70.24
91.196.50.33
93.113.125.11
95.213.177.122
95.213.177.123
95.213.177.124
95.213.177.125
95.213.177.126
```

Domains:

```
(kali㉿kali)-[~/Downloads]
$ grep -oP 'https?://[a-zA-Z0-9._/-]+' access-2.log | sort -u
http://104.236.11.102/
http://24x7-allrequestsallowed.com/
http://61.160.212.172
http://check.proxyradar.com/azenv.php
http://google.com/search
http://httpheader.net
http://login.yahoo.com/config/isp_verify_user
http://nmap.org/book/nse.html
https://104.236.11.102/ng12.zip
https://github.com/robertdavidgraham/masscan
https://proxyradar.com/
http://testp2.czar.bielawa.pl/testproxy.php
http://testp3.pospr.waw.pl/testproxy.php
http://testp4.pospr.waw.pl/testproxy.php
http://testp5.mielno.lubin.pl/testproxy.php
http://www.google.com/bot.html
```

User Accounts:

1. /admin/
2. /admin/i18n/readme.txt
3. /login/
4. /CFIDE/administrator/

```
(kali㉿kali)-[~/Downloads]
$ grep -oP '(?=<GET |POST )/[A-Za-z0-9._/-]*(?:login|admin|user|account|password|auth|wp-admin|phpmyadmin)[a-zA-Z0-9._/-]*' access-2.log | sort -u
/admin/
/admin/i18n/readme.txt
/CFIDE/administrator/
/login/
```

File Names:

xmlrpc.php , testproxy.php , favicon.ico , httpheader.net , ng12.zip , panel.zip , readme.txt , robots.txt , and sitemap.xml

```
(kali㉿kali)-[~/Downloads]
$ grep -oP 'GET \K[^\s]+' access-2.log | grep -oP '^[^/]+\.[A-Za-Z0-9]+$' | sort -u
favicon.ico
httpheader.net
ng12.zip
panel.zip
readme.txt
robots.txt
sitemap.xml
testproxy.php
xmlrpc.php
```

LOG FILE INFORMATION

- File Name: apache_logs
- File Type: ASCII text
- File Size: 2.3MB
- Source System:
- Source IP Address:
- Log Location: /home/kali/Downloads/apache_logs
- Hash Value : **SHA256hash**

f15c31e905f86c7b4b6ab44aee74d0a2086dce89f010187d983edea7ef0364ef

C. INITIAL OBSERVATIONS

- Number of Lines: 10,000
- Log Format: Combined Log Format (Apache/Nginx)
- Delimiter Used: Double quote character (")
- Time Range Covered: 3 days 11 hours
- Primary Fields Identified: 9 fields which are listed below:
 - ❖ Client IP Address
 - ❖ Remote Logname
 - ❖ Authenticated User
 - ❖ Timestamp

- ❖ Request Line (Method, Path, Protocol)
- ❖ HTTP Status Code
- ❖ Response Size
- ❖ Referer
- ❖ User-Agent

D. ANALYSIS OBJECTIVES

What am i looking for?

- ☒ Status code and response analysis
- ☒ Unauthorised Access
- ☐ Malware Activity
- ☒ Network Scanning
- ☒ Web Attacks
- ☐ Data Exfiltration
- ☐ Other: Suspicious User Activity

E. COMMANDS EXECUTED

Command	Purpose	Result
<code>wc -l apache_logs</code>	To count the number of lines in the file	10,000 lines
<code>awk '{print \$1}' apache_logs sort -u wc -l</code>	Unique IP addresses	1,753 unique IP's
<code>awk '{print \$1}' apache_logs sort uniq -c sort -rn head -1</code>	IP addresses that has more traffic	66.249.73.135 with 482 requests
<code>awk -F'"' '{print \$2}' apache_logs awk '{print \$2}' sort uniq -c sort -rn head -1</code>	Most visited URL	/favicon.ico with 807 requests
<code>awk -F'"' '{split(\$3,a," "); print a[1]}' apache_logs grep -c '^200\$'</code>	Total count of 200 responses	9,126 requests
<code>head -n 1 apache_logs grep -oP '\[.*?\]'</code>	Time range started	17/May/2015 10:05:03 +0000
<code>tail -n 1 apache_logs grep -oP '\[.*?\]'</code>	Time range ended	20/May/2015 21:05:15 +0000

1. How many lines are there in the log file in total?

```
(kali㉿kali)-[~/Downloads]
$ wc -l apache_logs
10000 apache_logs
```

2. How many unique IP's are there in the log file?

```
(kali㉿kali)-[~/Downloads]
$ awk '{print $1}' apache_logs | sort -u | wc -l
1753
```

3. What is the IP address from which the server got more traffic from?

```
(kali㉿kali)-[~/Downloads]
$ awk '{print $1}' apache_logs | sort | uniq -c | sort -rn | head -1
482 66.249.73.135
```

4. What is the URL which was most visited?

```
(kali㉿kali)-[~/Downloads]
$ awk -F' ' '{print $2}' apache_logs | awk '{print $2}' | sort | uniq -c | sort -rn | head -1
807 /favicon.ico
```

5. What is the total count of requests which got a 200 response?

```
(kali㉿kali)-[~/Downloads]
$ awk -F' ' '{split($3,a," "); print a[1]}' apache_logs | grep -c '^200$'
9126
```

F. FINDINGS:

1. Two hundred and thirteen HTTP 404 errors were recorded indicating active directory and resource enumeration across the server .
2. Two hundred and thirteen HTTP 404 errors were recorded indicating active directory and resource enumeration across the server, the highest 404 count across all three log files

G. INDICATOR OF COMPROMISE (IoCs)

Suspicious IP Addresses: 66.249.73.135 , 129.121.176.228 , 129.121.176.228 , 183.91.14.219 , 186.93.16.149 , 186.93.30.106 , 195.250.34.144 , 199.102.67.16 , 212.90.148.107 , 67.215.172.14

```
(kali@kali)-[~/Downloads]
$ grep -E 'CONNECT|\\\\x|admin|xmlrpc|masscan|zgrab' apache_logs | awk '{print $1}' | sort -u
129.121.176.228
130.185.72.6
144.76.194.187
183.91.14.219
186.93.16.149
186.93.30.106
188.165.243.45
192.185.81.134
192.185.83.181
195.250.34.144
198.245.61.43
199.102.67.16
199.116.117.212
199.168.96.66
199.189.248.95
201.242.142.135
212.90.148.107
216.14.208.102
216.150.76.218
217.26.210.20
46.28.105.80
50.87.144.128
5.9.143.150
62.24.122.25
65.55.213.73
66.147.244.126
66.249.73.135
67.215.172.14
74.208.16.115
81.169.144.135
85.19.71.167
91.236.75.25
95.78.54.93
98.130.2.118
(kali@kali)-[~/Downloads]
```


Domains:

```
(kali@kali)-[~/Downloads]
$ grep -oP 'https?://[a-zA-Z0-9._-]+' apache_logs | grep -v 'semicomplete\feedparser\google\bing\yahoo\lw3\.org' | sort -u | head -20
http://ahrefs.com/robot/
http://antonio-zabila.blogspot.com.es/2011/06/click-automaticos.html
http://askubuntu.com/questions/2316/how-can-i-map-a-character-to-a-key-combination
http://askubuntu.com/questions/88948/control-mouse-pointer-with-keyboard-without-a-number-pad
http://avtloads.net/
http://blackwitchcraft.ru/
http://blog.andreas-haerter.com/2012/02/05/how-to-start-puppet-system-config-links-resources-tutorials
http://blog.float.tw/2012/06/xdotool.html
http://blog.sleeplessbeastie.eu/2013/01/21/how-to-automate-mouse-and-keyboard/
http://blog.youxu.info/category/keyboard/
http://bruteforce.gr
http://codebright.wordpress.com/2010/06/27/sendkeys-in-linux/
http://community.spiceworks.com/topic/269270-how-can-i-pull-cpu-temperature-through-powershell-scripting
http://danceuniverse.ru/
http://den.im/
http://de.wikipedia.org/wiki/Xvfb
http://doc.ubuntu-fr.org/xbindkeys
http://dracorp.one.pl
http://en.wikipedia.org/wiki/Xvfb
http://es.wikipedia.org/wiki/Xvfb
```

User Accounts:

```
(kali@kali)-[~/Downloads]
$ grep -oP '(?<GET|POST)/[A-Za-z0-9._-]*(?:login|admin|user|account|password|auth|wp-admin|phpmyadmin)[a-zA-Z0-9._-]*' apache_logs | sort -u
/administrator/
/administrator/index.php
/admin.php
/articles/openldap-with-saslauthd/
/blog/geekery/boost-xpressive-supports-user-defined-assertions.html
/blog/geekery/zsh-disable-cd-username.html
/blog/tags/sysadmin
/blog/wp-admin/
/user/register
/wordpress/wp-admin/
/wp-admin/
/wp-login.php
/wp/wp-admin/
```

File Names:

```
(kali㉿kali)-[~/Downloads]
$ grep -oP 'GET \K[^\\s]+' apache_logs | grep -oP '^[^/?]+\.(php|zip|txt|sh|py|exe|bin)$' | sort -u | head -20
admin.php
albumcover.py
bingerpc.py
configlib.py
dcache.sh
dxargs.py
fastest_site.py
fastest_sites-20110317.py
fastest_sites.py
gterm.sh
happyfoo.py
index.php
java-ssl-debug-last-request.txt
java-ssl-debug.txt
json.txt
lumberjack-0.3.0.exe
main.py
migratemusic.py
mkgemrpms.sh
NERD_commenter.txt
```